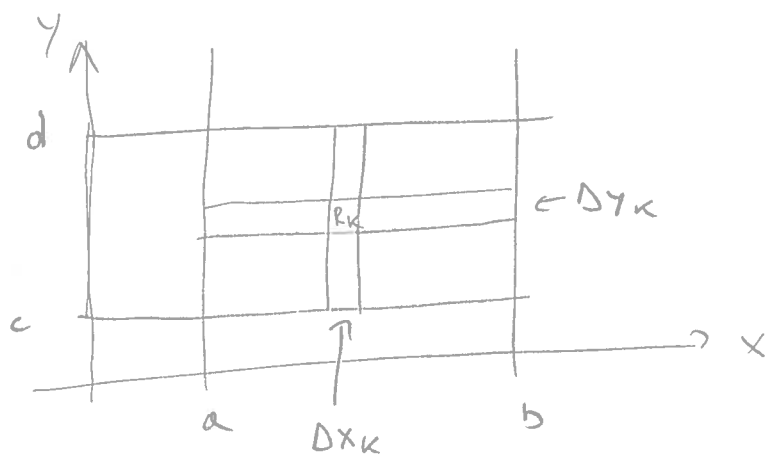


§ 13.1 Double integrals.

Defn $f(x, y)$ defined on a rectangle R is integrable if $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^m f(\bar{x}_k, \bar{y}_k) \Delta A_k$ exists

We let
$$\iint_R f(x, y) dA = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^m f(\bar{x}_k, \bar{y}_k) \Delta A_k$$



$$R = [a, b] \times [c, d]$$

Area of R_k is

$$\Delta A_k = \Delta x_k \cdot \Delta y_k$$

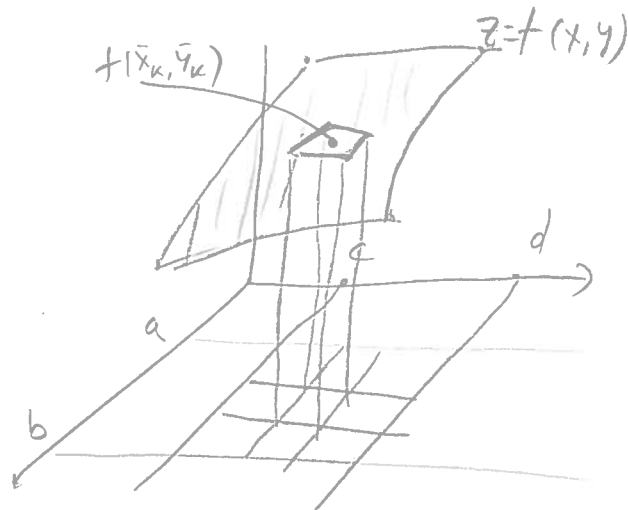
$$(\bar{x}_k, \bar{y}_k) \in R_k$$

So, for any $\epsilon > 0$ there is a $\delta > 0$ s.t.

if all Δx_k & Δy_k are less than δ , then for any

$(\bar{x}_k, \bar{y}_k) \in R_k$ we have

$$\left| L - \sum f(\bar{x}_k, \bar{y}_k) \Delta A_k \right| < \epsilon$$



Thm If f is bounded on a rectangle $R = [a, b] \times [c, d]$ & f is continuous (except on finitely many curves) then f is integrable on R .

So $\iint_{[0,1] \times [2,3]} e^{\cos(x+y)} \cdot \sin(x-y) dA$ exists.

But $\iint_{[0,1] \times [0,1]} \frac{x}{y} dA$ may not exist.

We have $\iint_R k f(x, y) dA = k \iint_R f(x, y) dA$

$$\iint_R [f(x, y) + g(x, y)] dA = \iint_R f(x, y) dA + \iint_R g(x, y) dA$$

$$R_1 \cap R_2 = \emptyset \quad \iint_{R_1 \cup R_2} f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$$

$R_1 \mid R_2$

$$f(x, y) \leq g(x, y) \quad \forall (x, y) \in R \Rightarrow \iint_R f(x, y) dA \leq \iint_R g(x, y) dA$$

$$\iint_R 1 dA = \text{area of } R$$

proof $\sum f(\bar{x}_k, \bar{y}_k) \Delta A_k = \sum \Delta A_k = \text{Area of } R \quad \square$

$$\iint_R k dA = k \cdot (\text{area of } R)$$

Ex Let $f(x,y) = \lfloor x \rfloor \cdot \lfloor y \rfloor$ where $\lfloor x \rfloor = \text{biggest integer } \leq x$

$$\begin{aligned} \text{so } \lfloor 1.5 \rfloor &= 1 \\ \lfloor 2 \rfloor &= 2 \\ \lfloor -3.1 \rfloor &= -4 \end{aligned}$$

compute $\iint_{[0,2] \times [0,2]} f(x,y) dA$

2	0	1
1	0	0
0	1	2

$$= \iint_{[0,1] \times [0,1]} 0 \cdot 0 dA + \iint_{[0,1] \times [1,2]} 0 \cdot 1 dA + \iint_{[1,2] \times [0,1]} 1 \cdot 0 dA + \iint_{[1,2] \times [1,2]} 1 \cdot 1 dA = 0 + 0 + 0 + 1 = 1$$

Ex Show that $9 \leq \iint_{[0,2] \times [0,2]} xy + 2 dA \leq 17$

2	R_3	R_4
1	R_1	R_2
0	1	2

$$2 \leq xy + 2 \leq 3 \text{ on } R_1$$

$$2 \leq xy + 2 \leq 4 \text{ on } R_2 \text{ \& } R_3$$

$$3 \leq xy + 2 \leq 6 \text{ on } R_4$$

$$9 = 2 + 2 + 2 + 3 \leq \iint_R xy + 2 \leq 3 + 4 + 4 + 6 = 17$$

Homework: (1) compute $\iint_{[1,4] \times [0,2]} f(x,y) dA$ $f(x,y) = \begin{cases} 2 & 1 \leq x \leq 4 \quad 0 \leq y \leq 1 \\ 3 & 1 \leq x \leq 3 \quad 1 \leq y \leq 2 \\ 1 & 3 \leq x \leq 4 \quad 1 \leq y \leq 2 \end{cases}$

(2) Divide $R = [0,6] \times [0,4]$ in 6 equal squares and approximate $\iint_R f(x,y) dA$ by $\sum_{k=1}^6 f(\bar{x}_k, \bar{y}_k) \Delta A_k$ for

a) $f(x,y) = 12 - x - y$ b) $f(x,y) = x^2 + 2y^2$ c) $f(x,y) = e^{xy}$